

Oleksiy Nedobiychuk

Software Engineer | Python/Django | TypeScript | IoT & Data Platforms

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🌐 <http://nedo.im/blog/>

📍 Pisa, Italy

Education

- **M.Sc. in Computer Science (AI curriculum), University of Pisa** [2023 – Present] 109/110(e).
- **B.Sc. in Computer Science, University of Pisa** [2019 – 2023] 106/110.

Relevant Experience

Software Engineer at TD Group Italia S.r.l., Vecchiano, Italy [Jul. 2025 – Present]:

- Developed full-stack **Django/MySQL** platforms for agritech and healthcare IoT projects (VITE, Smart-Care, EyeDos), covering device management, role-based admin workflows, and production-oriented data models.
- Built **REST API**/admin workflows and **Keycloak**/OIDC role-based access, integrating multi-user authorization patterns across Django and Angular/Spring systems.
- Implemented **MQTT**-driven ingestion and real-time update flows with **Channels/Redis**; created **Highcharts/ECharts** dashboards for telemetry, model outputs, and operational data.
- Built public **Django**/Tailwind project sites for HeartSentry and VITE, including content pages, contact/privacy flows, and deployable static/media configuration.

Full-Stack Developer at TAI Solutions S.r.l., Pisa, Italy [Sept. – Nov. 2022]:

- Built a **Spring** microservice and **Angular** application for tenant/user management, integrating **Keycloak** authentication and authorization across multiple clients.

Projects

Professional Cycling Analysis – Data Mining

[GitHub](#)

- Cleaned and engineered cycling datasets with imputation, outlier detection, and correlation analysis; applied K-means/DBSCAN clustering and trained a neural network to predict rider performance with 90% test accuracy.

Neural Network for Regression/Classification

[GitHub](#)

- Built a neural network from scratch in **Python/NumPy** with backpropagation, Nesterov momentum, L1/L2 regularization, and grid-search tuning; reached 0.01 MSE on regression and 95% accuracy on classification.

ProbLog Based AI Agent

[GitHub](#)

- Built a probabilistic AI agent for a 2D fighting game, modeling uncertainty with a ProbLog knowledge base and dynamic probability estimation; outperformed MCTS, MinMax, and Prolog baselines in simulation.

Web Application – Full-Stack Development

[Live](#)

- Developed and deployed a **Svelte**/WebGL2 frontend with a **Django**/PostgreSQL backend on Debian using Nginx and Apache, including database design and production server configuration.

Skills

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|---------------------|---|
| Programming | • Proficient: Python, TypeScript/JavaScript. Familiar: C++, C, Rust. Learning: Go. |
| Backend/Data | • Django, Spring, REST APIs, MySQL, PostgreSQL, MongoDB, Redis, Keycloak/OIDC. |
| IoT/DevOps | • MQTT, Docker, Linux (Ubuntu, Arch), Shell scripting, Git, Nginx/Apache. |
| Frontend | • Angular, Svelte, TypeScript, Highcharts, ECharts, WebGL2, responsive UI development. |
| Tools | • Vim, Copilot, ChatGPT; ML/DS familiarity with PyTorch, Scikit-learn, Pandas, NumPy. |
| Languages | • Fluent in English, Italian, and Ukrainian. |